A CASE STUDY ON THE COMPARISON OF OSI AND TCP/IP

REFERENCE MODELS WITH RESPECT TO LAYERS, FUNCTIONS,

PROTOCOLS, AND REAL-WORLD IMPLEMENTATION

A Research Report

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Abstract

Computer networking today rests on two layered reference architectures: the seven-layer Open Systems Interconnection (OSI) model published by the International Organization for Standardization, and the four-layer Transmission Control Protocol/Internet Protocol (TCP/IP) model that grew out of the United States Department of Defense's ARPANET project [1][8]. Although the two models were developed for different purposes and at different points in the evolution of networking, they are routinely discussed side by side because they describe overlapping aspects of the same problem: how two independent computer systems can exchange data reliably across heterogeneous hardware and software. This report presents a case study comparing the OSI and TCP/IP reference models across four dimensions — their layer architecture, the functions performed at each layer, the protocols associated with each layer, and their real-world implementation in modern networks. The study traces the historical development of both models, analyses their structural and functional differences, reviews how security is distributed across their layers, and considers their continued relevance in the era of cloud computing, software-defined networking, and the impending transition to IPv6. The report concludes that while the OSI model remains the dominant pedagogical and analytical reference framework, the TCP/IP model is the model that is actually implemented in production networks and the modern Internet.

Contents

1. Introduction 3
1.1 Background of Computer Networking 3
1.2 Need for Standardized Networking Models 3
1.3 Problem Statement 4
1.4 Research Objectives and Questions 4
1.5 Scope and Organization of the Report 4
2. Historical Development and Literature Review 5
2.1 Evolution of Computer Networks 5
2.2 Development of ARPANET and TCP/IP 5
2.3 Development of the OSI Reference Model 6
2.4 Review of Previous Research 6
3. Research Methodology 7
4. Overview of the OSI Reference Model 8
4.1 Objectives and Architecture 8
4.2 The Seven Layers 8
4.3 Advantages and Limitations 9
5. Overview of the TCP/IP Reference Model 10
5.1 Architecture and History 10
5.2 The Four Layers 10
5.3 Advantages and Limitations 11
6. Comparative Analysis of OSI and TCP/IP Models 12
6.1 Architectural Comparison 12
6.2 Layer Mapping 12
6.3 Functional and Protocol Comparison 13
6.4 Performance and Scalability 13
7. Protocol Analysis 14
8. Data Communication Process 15
9. Network Security Analysis 16
10. Modern Applications and Emerging Technologies 17
11. Case Studies 18
12. Advantages and Limitations: Overall Evaluation 19
13. Future Scope of Network Architecture 19
14. Conclusion 20
References 21

1. Introduction

1.1 Background of Computer Networking

Modern computing depends on the ability of independent, often dissimilar, machines to exchange information accurately and efficiently. In the early decades of computing, manufacturers built proprietary networking systems such as IBM's Systems Network Architecture (SNA) and Digital Equipment Corporation's DECnet, each optimized for its own hardware and incompatible with equipment from other vendors [3]. As organizations began connecting machines from different manufacturers, this fragmentation became a serious obstacle to growth. The response of the networking community was to develop layered reference models: conceptual frameworks that decompose the complex task of network communication into a stack of smaller, well-defined problems, each solved independently at its own layer [1][4].

1.2 Need for Standardized Networking Models

A reference model provides a common vocabulary and a common set of expectations that allow hardware vendors, software developers, and network engineers to build interoperable systems without needing to understand every other layer of the stack in detail [4][7]. Layering also simplifies troubleshooting, because a fault can usually be isolated to a single layer, and it accelerates innovation, because a layer can be redesigned or replaced without disturbing the layers above or below it [7]. Two layered models dominate the study and practice of networking today: the OSI reference model, produced by the International Organization for Standardization (ISO), and the TCP/IP model, which emerged from United States Department of Defense-funded research on the ARPANET [1][8][9].

1.3 Problem Statement

Despite being taught together in almost every networking curriculum, the OSI and TCP/IP models differ substantially in their origin, structure, and real-world adoption. Students and practitioners are frequently confused about why a seven-layer model that is rarely implemented in its complete form (OSI) continues to serve as the principal teaching and analytical reference, while a four-layer model that was never formally standardized as a reference model in the way OSI was (TCP/IP) underlies almost all of the world's actual data networks, including the Internet [8][10]. This report addresses that problem by systematically comparing the two models.

1.4 Research Objectives and Questions

The objectives of this case study are to:

- Trace the historical circumstances that produced the OSI and TCP/IP reference models.
- Describe the layer architecture, functions, and associated protocols of each model.
- Compare the two models with respect to structure, protocol mapping, security distribution, and performance characteristics.
- Examine how each model is expressed in modern technologies such as cloud computing, software-defined networking, and IPv6.
- Draw conclusions on the practical relevance of each model in contemporary network design.

Correspondingly, the report is organized around the following research questions: (i) How do the layer structures of OSI and TCP/IP correspond to one another? (ii) What functional and protocol-level differences exist between the two models? (iii) How is network security distributed across the layers of each model? (iv) What is the continuing relevance of each model to modern and emerging network architectures?

1.5 Scope and Organization of the Report

The scope of this report is limited to a conceptual and comparative analysis of the two reference models; it does not attempt an exhaustive protocol-level specification of either stack, nor does it evaluate any single vendor's product implementation. The remainder of the report is organized as follows: Chapter 2 reviews the historical development of both models; Chapter 3 outlines the research methodology; Chapters 4 and 5 describe the OSI and TCP/IP models individually; Chapter 6 presents the comparative analysis; Chapter 7 examines the protocols associated with each layer; Chapter 8 describes the data communication process; Chapter 9 analyses network security; Chapter 10 discusses modern applications; Chapter 11 presents case studies; Chapters 12 and 13 discuss advantages, limitations, and future scope; and Chapter 14 concludes the report.

2. Historical Development and Literature Review

2.1 Evolution of Computer Networks

Computer networking evolved from single-vendor, hardware-specific communication schemes in the 1960s and 1970s toward increasingly open, standardized systems. Early wide-area networking research was driven largely by government-funded projects aimed at resource sharing between geographically dispersed research computers, the most influential of which was ARPANET [9][12].

2.2 Development of ARPANET and TCP/IP

The Advanced Research Projects Agency (ARPA, later DARPA) of the U.S. Department of Defense began work on ARPANET in the late 1960s, and the network became operational in 1969, initially connecting four host computers using the Network Control Program (NCP) [12][16]. As the network grew, NCP's limitations became apparent, and DARPA researchers Vinton Cerf and Robert Kahn developed a new protocol, first documented in RFC 675 (1974) as the Transmission Control Program [10]. Around 1978 this monolithic protocol was split into two layers — the Transmission Control Protocol (TCP) and the Internet Protocol (IP) — and formally specified in RFCs 791, 792, and 793 (1981) [10]. The U.S. Department of Defense mandated TCP/IP as the standard for military networking in March 1982, and ARPANET completed its transition from NCP to TCP/IP on 1 January 1983 [12][16]. The resulting protocol suite is described, from lowest to highest layer, as the Link, Internet, Transport, and Application layers in RFC 1122 (1989) [8][17]. Because TCP/IP was defined descriptively — as a specification of protocols already in deployment — rather than prescriptively, it is often distinguished from the OSI Reference Model, which was designed in advance of implementation [8][13].

2.3 Development of the OSI Reference Model

Recognizing the same interoperability problems that motivated the ARPANET work, ISO began developing a vendor-neutral reference model in 1977, later merging its effort with a parallel initiative from the International Telegraph and Telephone Consultative Committee (CCITT) [3]. The result, the Open Systems Interconnection Reference Model, was published in 1984 as ISO 7498 and later refined as ISO/IEC 7498-1:1994 [1][2][4]. Unlike TCP/IP, OSI was developed as a prescriptive architecture: a blueprint intended to guide the design of new protocols, rather than a description of an existing implementation [8][13]. Proprietary architectures such as DECnet were subsequently re-engineered to comply with OSI, illustrating the pressure that the international standard placed on vendor-specific systems even though OSI's own protocol suite saw limited real-world deployment [3].

2.4 Review of Previous Research

The literature on OSI and TCP/IP spans standards documents, vendor white papers, and pedagogical and security-oriented analyses. Padlipsky's RFC 871 (1982) offers an early first-person account contrasting the ARPANET Reference Model with the ISO Reference Model, arguing that the two were built on different design philosophies [13]. Subsequent literature, including vendor and industry publications from AWS, BMC, Imperva, and Corero, has focused on describing OSI's seven layers as a teaching and troubleshooting framework [1][4][6][7], while networking-security literature from sources such as Tripwire, Exabeam, and DestCert has examined how the same seven layers map onto categories of network attack and defense [14][15][17]. More recent research addresses how both models interact with software-defined networking (SDN) and the IPv4-to-IPv6 transition [19][20][22]. This report synthesizes these strands into a single, structured comparison.

3. Research Methodology

3.1 Research Approach

This report follows a qualitative, descriptive-comparative research design. Rather than collecting primary experimental data, the study synthesizes secondary source material — standards documents, RFCs, vendor technical documentation, and peer-reviewed literature — into a structured comparison of the OSI and TCP/IP models.

3.2 Data Collection Methods and Sources of Information

Source material was drawn from four categories: (i) foundational standards and requests for comments, including RFC 791, RFC 1122, and ISO/IEC 7498-1 as referenced in secondary literature [2][8][10]; (ii) vendor and industry technical publications from organizations such as AWS, BMC, Imperva, and Corero [1][4][6][7]; (iii) network-security literature examining layer-specific threats and countermeasures [14][15][16][17][18]; and (iv) academic and industry research on emerging technologies such as SDN, NFV, and IPv6 migration [19][20][21][22].

3.3 Analysis Framework

The analysis framework compares the two models along four axes identified in the research questions: architectural structure (number and boundaries of layers), functional responsibility (what each layer does), protocol content (which protocols are associated with each layer), and real-world implementation (how each model manifests in deployed systems, including security architecture and emerging technologies). A layer-mapping table (Chapter 6) and function/protocol tables (Chapters 4, 5, and 7) are used throughout to keep the comparison systematic.

3.4 Limitations of the Study

As a desk-based comparative study, this report does not include laboratory measurement of network performance, and its comparative claims about efficiency and scalability rely on characteristics reported in the secondary literature rather than original benchmarking. In addition, because OSI's own protocol suite was never widely deployed, the report necessarily discusses OSI primarily as a conceptual and pedagogical model while discussing TCP/IP as both a conceptual model and a deployed protocol suite [8][13].

4. Overview of the OSI Reference Model

4.1 Introduction to OSI Architecture and Objectives

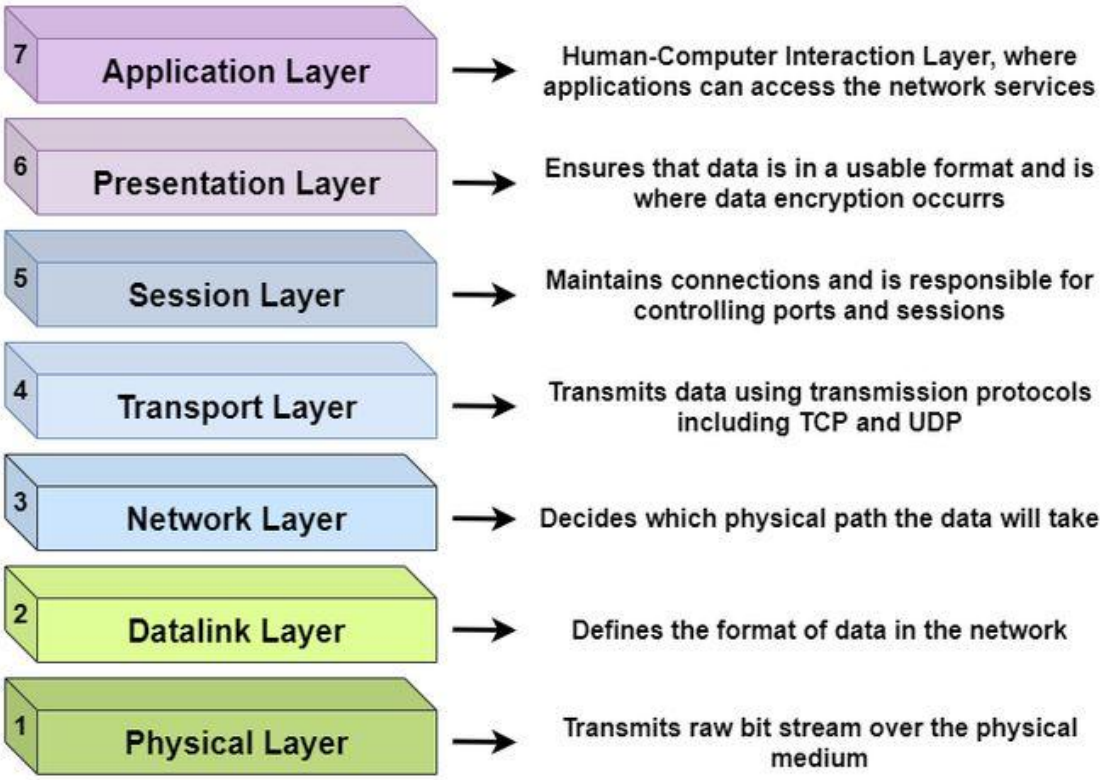
The OSI model is a conceptual framework that divides network communication into seven abstract layers, each performing specific functions and offering services to the layer above it while relying on the services of the layer below it [1][7]. The lowest three layers are primarily concerned with moving traffic through the physical network toward an end system, while the top four layers operate within the end systems themselves to complete the communication process [2]. Layers 1 through 4 together are sometimes described as the transport system, and layers 5 through 7 as the application system [3]. The model's central objective is to allow any two systems that correctly implement the same layer interfaces to interoperate, regardless of the underlying hardware or software vendor [4][7].

4.2 The Seven Layers

Table 4.1 summarizes the seven OSI layers, their principal function, and representative protocols or technologies associated with each layer.

Layer	Name	Principal Function	Example Protocols/Technologies
7	Application	Provides network services directly to end-user applications	HTTP, FTP, SMTP, DNS
6	Presentation	Translates, encrypts, and compresses data for the application layer	TLS/SSL, JPEG, ASCII/EBCDIC
5	Session	Establishes, manages, and terminates sessions between applications	NetBIOS, RPC, PPTP
4	Transport	End-to-end error recovery, flow control, and segmentation	TCP, UDP
3	Network	Logical addressing and routing of packets between networks	IP, ICMP, OSPF
2	Data Link	Framing, physical addressing, and error detection on a single link	Ethernet, PPP, Wi-Fi (802.11)
1	Physical	Transmission of raw bits over a physical medium	Cabling, fiber optics, Bluetooth

Each layer's boundary is defined so that the units within a layer can, in principle, be exchanged without requiring changes in adjacent layers [3]. Data passed from an application descends through each layer on the sending system, gaining a layer-specific header (and, at the data link layer, a trailer) at every step — a process known as encapsulation — before being transmitted; the receiving system reverses the process through decapsulation [2][7].



4.3 Advantages and Limitations of the OSI Model

The principal advantages commonly attributed to OSI are its use as a universal teaching language for networking, its fine-grained isolation of function which simplifies fault diagnosis, and its support for modular, vendor-independent innovation at each layer [4][6][7]. Its principal limitation is that the complete seven-layer OSI protocol suite was never widely adopted in production networks; the model is used today chiefly as an analytical and pedagogical reference rather than as an implemented stack, and several of its layer boundaries — particularly between the session, presentation, and application layers — are difficult to observe cleanly in real protocols [8][17].

5. Overview of the TCP/IP Reference Model

5.1 Introduction to TCP/IP Architecture and History

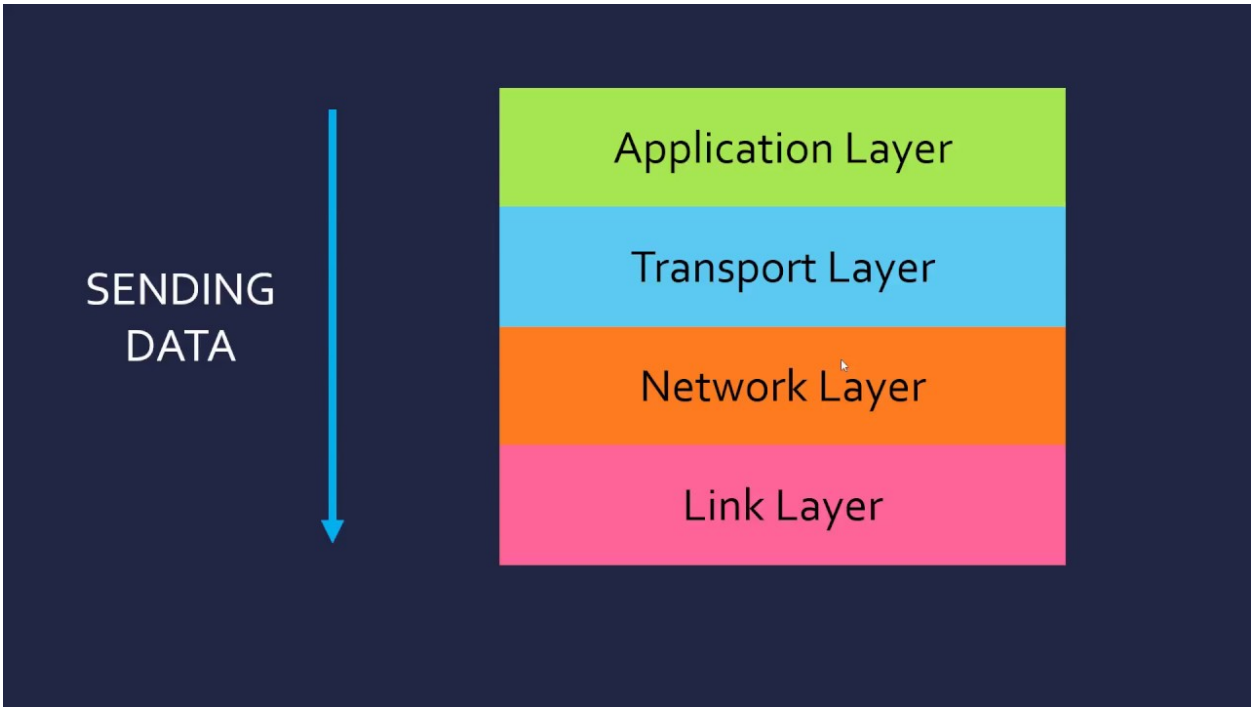
The TCP/IP model, sometimes called the Internet model or the DoD model, is a four-layer description of the protocol suite that DARPA developed in the 1970s to succeed ARPANET's original Network Control Program [8][9]. Unlike OSI, TCP/IP was not designed in advance as an abstract reference architecture; it was formally documented, in RFC 1122 (1989), as a description of protocols that were already running in production on ARPANET and its successor networks [8][17]. This descriptive origin is the reason authors such as Padlipsky treat TCP/IP and OSI as products of fundamentally different design philosophies, even though they are frequently placed side by side in comparative diagrams [13].

5.2 The Four Layers

Table 5.1 summarizes the four TCP/IP layers, their function, and representative protocols.

Layer	Principal Function	Example Protocols
Application	Provides process-to-process services and end-user protocols	HTTP, FTP, SMTP, DNS, SSH
Transport	Provides host-to-host communication, reliability, and flow control	TCP, UDP
Internet	Provides logical addressing, routing, and packet forwarding across networks	IP (IPv4/IPv6), ICMP, ARP
Link (Network Access)	Provides delivery of frames within a single physical/local network	Ethernet, Wi-Fi, PPP

The Application layer of TCP/IP collapses the OSI Application, Presentation, and Session layers into a single layer, on the reasoning that these functions are typically implemented within application software itself rather than as separate network services [17]. The Internet layer maps closely to the OSI Network layer, and the Transport layer maps directly to OSI's Transport layer of the same name [17]. Some authors further subdivide the Link layer into separate physical and data-link components, producing a five-layer variant, but the IETF has never altered the four-layer structure defined in RFC 1122 [17].



5.3 Advantages and Limitations of the TCP/IP Model

TCP/IP's principal advantage is that it describes the protocol suite that is actually deployed across the modern Internet, giving it unmatched real-world relevance, simplicity, and a long track record of scalability [8][16]. Its principal limitation, from an analytical standpoint, is that its coarser four-layer structure is less granular than OSI, which can make it harder to isolate the precise function or security control associated with a given layer

boundary — for example, the Application layer must account for session, presentation, and true application concerns simultaneously [17][22].

6. Comparative Analysis of OSI and TCP/IP Models

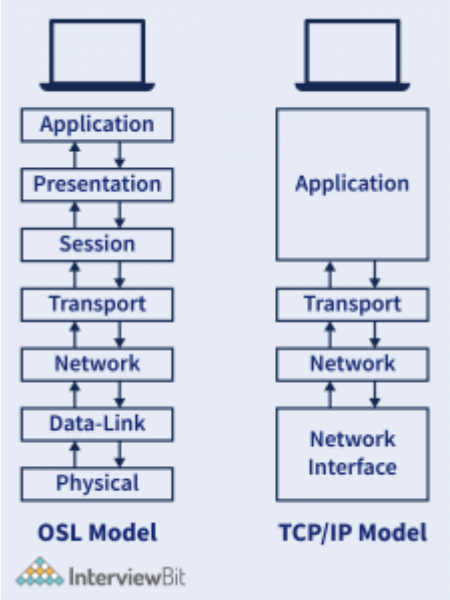
6.1 Architectural Comparison

The most immediately visible difference between the two models is structural: OSI defines seven layers while TCP/IP defines four [1][8]. This difference in granularity reflects a difference in purpose. OSI was designed prescriptively, as a reference against which new protocols could be measured and designed, and ISO therefore had the freedom to separate concerns — such as session management and data presentation — into distinct layers even where no single deployed protocol occupied that layer alone [8][13]. TCP/IP, in contrast, was documented descriptively, capturing the structure of a protocol suite that had already been implemented and deployed, and its layer boundaries therefore reflect the protocols that existed rather than an idealized separation of concerns [8][17].

6.2 Layer Mapping Between OSI and TCP/IP

Table 6.1 maps the OSI layers onto their nearest TCP/IP equivalents.

OSI Layer	Nearest TCP/IP Layer	Notes
7. Application	Application	TCP/IP merges Application, Presentation, and Session into one layer [17]
6. Presentation	Application	Encryption/compression handled within application libraries (e.g., TLS) [17][22]
5. Session	Application	Session management left to the application or transport protocol [17]
4. Transport	Transport	Direct one-to-one correspondence (TCP, UDP) [17]
3. Network	Internet	Direct correspondence; both centre on IP addressing/routing [17]
2. Data Link	Link	TCP/IP's Link layer covers framing and local addressing [17]
1. Physical	Link	TCP/IP does not formally separate a physical sub-layer [17]



6.3 Functional and Protocol Comparison

Functionally, both models perform the same essential tasks — physical transmission, addressing, routing, reliable delivery, and application service provision — but distribute responsibility differently. Where OSI assigns encryption/compression to a dedicated Presentation layer and session establishment to a dedicated Session layer, TCP/IP delegates both responsibilities to protocols operating at or above the Transport layer, such as TLS, which is frequently described as sitting 'between' the Transport and Application layers of the TCP/IP stack even though it is conceptually an OSI Layer 6 function [15][18][22]. In terms of protocol content, the two models converge heavily at the lower and middle layers — both rely on the same physical and data-link technologies (Ethernet, Wi-Fi), the same internetworking protocol (IP), and the same transport protocols (TCP, UDP) — because in practice only one protocol suite is deployed, described by two different conceptual lenses [8][17].

6.4 Performance and Scalability Comparison

Because the same underlying protocols (IP, TCP, UDP, Ethernet) support both descriptions, neither model imposes an inherent performance cost by itself; performance and scalability are determined by the protocols and implementations in use, not by which conceptual model is used to describe them [8][17]. However, the TCP/IP model's coarser four-layer structure is often credited with easier and cheaper practical implementation, since fewer formal layer boundaries need to be crossed, which is one reason TCP/IP became the protocol suite of the commercial Internet while OSI's own protocol suite saw far more limited deployment [8][16]. OSI's finer granularity, meanwhile, provides scalability of a different kind: it scales analytically, allowing new technologies (such as SDN control-plane separation) to be discussed and located precisely at a specific layer boundary [19][22].

7. Protocol Analysis

7.1 OSI Protocol Suite

ISO originally paired the OSI reference model with its own protocol suite, including X.400 (messaging), X.500 (directory services), and the connection-oriented network protocol CLNP, intended to compete directly with the ARPANET-derived TCP/IP suite [3][13]. With the exception of a few surviving elements — such as the IS-IS routing protocol, still used inside some large ISP backbones — this native OSI protocol suite was largely abandoned in favour of TCP/IP-based protocols during the 1990s [3].

7.2 TCP/IP Protocol Suite

The TCP/IP suite that dominates today's networks includes the Internet Protocol (IPv4/IPv6) for addressing and routing, the Transmission Control Protocol (TCP) for reliable, connection-oriented delivery, the User Datagram Protocol (UDP) for lightweight, connectionless delivery, and a large family of Application-layer protocols including HTTP/HTTPS, DNS, SMTP, and SSH [8][17]. Because these protocols were standardized through the open, iterative RFC process managed by the IETF, the suite has been able to evolve incrementally — for example, extending IP from a 32-bit to a 128-bit address space in IPv6 — without requiring a wholesale replacement of the model [10][17][21].

7.3 Reasons Behind TCP/IP Dominance

Several factors explain why TCP/IP, rather than OSI's native suite, became the protocol suite of the global Internet: TCP/IP was free, openly documented through RFCs, and already running in production on a growing research network (ARPANET) years before OSI's protocols were finalized; the U.S. government's 1982 mandate requiring DoD networks to use TCP/IP created a large installed base; and the comparative simplicity of a four-layer suite reduced the cost of vendor implementation relative to the more complex OSI stack [9][12][16]. By the time OSI protocols reached market readiness in the late 1980s, TCP/IP had already achieved substantial commercial and academic momentum, and the emergence of the World Wide Web on top of TCP/IP in the early 1990s cemented its position [3][8].

8. Data Communication Process

8.1 Data Encapsulation and Decapsulation

In both models, data produced by an application is passed down through the layers of the stack, and each layer adds its own control information — a process called encapsulation — before the resulting unit is transmitted across the physical medium [2][7]. At the receiving end, each layer strips off and interprets the corresponding control information as the data passes back up the stack, a process called decapsulation. This layered exchange allows each layer to operate as if it were communicating directly and privately with its peer layer on the remote system, even though the actual data path passes down through every layer on the sender and up through every layer on the receiver [4][7].

8.2 Protocol Data Units (PDUs)

As data descends the stack it is given different names at different layers, reflecting the form the control information takes at each stage. Table 8.1 lists the conventional naming.

OSI Layer	Protocol Data Unit (PDU)
Application/Presentation/Session	Data / Message
Transport	Segment (TCP) / Datagram (UDP)

OSI Layer	Protocol Data Unit (PDU)
Network	Packet
Data Link	Frame
Physical	Bits

8.3 Example: Web Page Request

A typical HTTPS request illustrates the process end to end. The browser's HTTP request is passed to the TLS library for encryption (Presentation-layer function), the resulting data is handed to TCP, which segments it and adds sequence and port information (Transport layer), TCP's segment is handed to IP, which adds source and destination addresses (Network/Internet layer), and the IP packet is handed to the Ethernet or Wi-Fi driver, which frames it with MAC addresses and a checksum before it is transmitted as bits (Data Link and Physical layers) [1][7][18]. The receiving web server reverses this sequence, and each layer's added header is used only by its peer layer on the remote host.

9. Network Security Analysis

9.1 Importance of Network Security in Layered Models

Because both models decompose communication into layers, they also provide a natural framework for reasoning about where security controls should be applied and where an attacker might target a given weakness. Defense-in-depth strategies typically apply distinct, complementary controls at multiple layers simultaneously, so that failure of one control does not compromise the entire communication path [15][17].

9.2 Security in OSI Layers

Security mechanisms map onto specific OSI layers: physical security (locks, shielding) applies at Layer 1; MAC-address filtering, VLAN segmentation, and link-layer encryption such as MACsec apply at Layer 2; IPsec, which provides packet authentication and encryption via its Authentication Header (AH) and Encapsulating Security Payload (ESP) components, applies at Layer 3; TCP/UDP-level protections and TLS, though more commonly discussed as a Presentation-layer (Layer 6) mechanism, secure Layer 4 traffic; and application firewalls, OAuth, and Kerberos operate at Layer 7 [15][16][18][23].

9.3 Security Mechanisms in TCP/IP

Within the TCP/IP model's four layers, security is distributed more coarsely: link-layer protections (MAC filtering, VLANs) sit within the Link layer; IPsec operates within the Internet layer; TLS/SSL secures the boundary between the Transport and Application layers, commonly described as 'HTTP over TLS over TCP' in the case of HTTPS; and authentication/authorization frameworks such as OAuth and Kerberos, together with application firewalls, operate within the Application layer [16][18][23].

9.4 Common Network Attacks by Layer

Table 9.1 lists representative attacks associated with each layer of the OSI model, together with common mitigations.

Layer	Representative Attacks	Common Mitigations
Physical	Cable tapping, hardware tampering, jamming	Physical access control, shielded cabling
Data Link	MAC spoofing, ARP poisoning, VLAN hopping	Port security, dynamic ARP inspection, VLAN segmentation
Network	IP spoofing, route injection, DDoS floods	Firewalls, IPsec, routing authentication, ACL filtering [20][17]
Transport	TCP SYN flooding, session hijacking	SYN cookies, rate limiting, TLS
Session/Presentation	Session hijacking, SSL/TLS downgrade or interception	Strong session tokens, certificate pinning, enforced modern TLS versions [17][23]
Application	SQL injection, cross-site scripting, credential attacks	Input validation, application firewalls, MFA

9.5 Security Challenges in Modern Networks

Virtualized and cloud-native environments introduce security challenges that cut across the traditional layer boundaries of both models: hypervisor and virtual-switch compromises threaten tenant isolation at the lower layers, while software-defined networking controllers introduce a new, logically centralized attack surface at the network/internet layer that has no clean analogue in either the original OSI or TCP/IP specifications [19][23]. Consequently, contemporary security practice increasingly layers additional controls — micro-segmentation, identity-based routing, and secure tunnelling protocols — on top of the classic layer-by-layer model [23].

10. Modern Applications and Emerging Technologies

10.1 Cloud Computing Networks

Cloud computing platforms rely on the same TCP/IP protocol suite as the rest of the Internet, but add virtualization layers — virtual switches, overlay networks, and multi-tenant IP address translation — on top of the Internet and Link layers to allow many isolated customer networks to share common physical infrastructure [19][20].

10.2 Software-Defined Networking (SDN) and NFV

Software-defined networking separates the network's control plane (which decides how traffic should be routed) from its data plane (which forwards traffic), centralizing control-plane logic in a software controller that can be reasoned about independently of the underlying Network/Internet layer hardware [19][22]. Network Function Virtualization (NFV) complements SDN by replacing dedicated hardware appliances — routers, firewalls, load balancers — with software functions running on commodity servers [19]. Research on integrating SDN with next-generation addressing, sometimes termed Software-Defined IPv6 (SoDIP6) networking, has shown that combining these two shifts can reduce operational cost and energy consumption while easing the technical burden of the IPv4-to-IPv6 transition [20][22].

10.3 IPv6, 5G, Edge Computing, and AI-Driven Networks

The exhaustion of the 32-bit IPv4 address space — which occurred progressively by region, beginning in Asia in 2011 and reaching North America and Africa by 2015-2017 — has made the transition to the 128-bit IPv6

addressing scheme, standardized by the IETF, an increasingly urgent element of Internet-layer evolution [21][22]. Fifth-generation (5G) mobile networks and edge computing extend both models' Application- and Transport-layer concerns toward much lower latency requirements, while machine-learning-based traffic engineering is increasingly used within SDN controllers to automate routing and security decisions that were previously performed manually at the Network layer [20][22].

11. Case Studies

11.1 ARPANET's Adoption of TCP/IP

ARPANET's transition from the Network Control Program to TCP/IP, completed on 1 January 1983 after a U.S. Department of Defense mandate the previous year, is the clearest historical case of a live, operational network migrating between competing internetworking architectures [12][16]. The case illustrates how an installed base of running infrastructure, combined with a policy mandate from a major institutional customer, can determine which reference architecture becomes dominant in practice, independent of which architecture is more formally elegant [9][13].

11.2 IPv4-to-IPv6 Migration as a Case in Internet-Layer Evolution

The ongoing, multi-decade migration from IPv4 to IPv6 illustrates how the Internet/Network layer of the TCP/IP and OSI models can be evolved without disturbing the layers above or below it. IPv6 was introduced by the IETF to resolve IPv4 address exhaustion by expanding the address field from 32 to 128 bits, while also simplifying the packet header — for example, by removing the checksum field that IPv4 required routers to recompute on every hop [21][35]. Despite these benefits, IPv6 adoption has proceeded slowly in many regions because dual-stack and tunnelling transition mechanisms add operational complexity, and because IPv4 and IPv6 addresses are not directly interoperable [20][21][22]. Research into combining IPv6 migration with SDN-based network management (SoDIP6) has been proposed specifically to reduce this transition cost for internet service providers in developing markets [20][22].

12. Advantages and Limitations: Overall Evaluation

Table 12.1 summarizes the overall comparative evaluation developed across Chapters 4 through 9.

Criterion	OSI Model	TCP/IP Model
Origin	Prescriptive; designed by ISO before implementation [8][13]	Descriptive; documented from an already-deployed suite [8][17]
Layers	Seven, finer-grained separation of concerns [1]	Four, coarser and simpler [8]
Real-world deployment	Native protocol suite largely abandoned; model used analytically [3][13]	Underlies the modern Internet in practice [8][16]
Teaching value	Preferred for isolating and naming specific functions [4][6][7]	Preferred for describing deployed systems directly [17]
Security granularity	Finer per-layer mapping of controls [15][18]	Coarser; several controls share the Application layer [16][18]

Criterion	OSI Model	TCP/IP Model
Adaptability	Standardization process slower and more formal [3][13]	Evolves incrementally via the open IETF RFC process [10][17]

Overall, the two models should be understood as complementary rather than competing descriptions of the same underlying communication problem: OSI offers analytical precision well suited to teaching, troubleshooting, and security-control mapping, while TCP/IP offers a faithful, simpler description of the protocol suite that is actually deployed across the Internet [4][8][17].

13. Future Scope of Network Architecture

13.1 Evolution Toward 6G, Pervasive IoT, and Quantum Networking

As networks continue to evolve toward sixth-generation (6G) mobile standards, pervasive Internet-of-Things deployments, and early quantum-communication research, both reference models are likely to remain in use as complementary lenses rather than being replaced outright: OSI's fine-grained layer separation will likely continue to serve as the reference framework for locating new security and management functions precisely, while the deployed protocol stack will continue to be described, in practice, using the simpler TCP/IP-oriented terminology already used across SDN, NFV, and IPv6 literature [19][20][22]. Continued growth in the number and diversity of connected devices is likely to keep the IPv4-to-IPv6 transition, and the broader move toward software-defined, virtualized network management, at the center of Internet-layer research for the foreseeable future [20][21][22].

14. Conclusion

14.1 Summary of Findings

This report has compared the OSI and TCP/IP reference models across their historical origins, layer architecture, functional responsibilities, protocol content, security distribution, and relevance to modern network technologies. The two models share a common purpose — enabling interoperable communication between independent systems — but differ in origin and granularity: OSI is a prescriptive, seven-layer standard published by ISO in 1984, while TCP/IP is a descriptive, four-layer model documenting the protocol suite that grew out of DARPA's ARPANET project and was formally defined in RFC 1122 in 1989 [1][8][10]. Layer-by-layer comparison shows that TCP/IP's single Application layer absorbs the functions that OSI separates into its Application, Presentation, and Session layers, while the two models converge closely at the Network/Internet and Transport layers [17]. Security responsibilities, protocol suites, and emerging technologies such as SDN, NFV, and IPv6 can all be mapped onto either model, though OSI's finer granularity tends to be preferred for security-control mapping and TCP/IP's simplicity for describing deployed infrastructure [15][19][22].

14.2 Closing Remarks

Ultimately, the persistence of both models reflects a genuine division of labour between them: OSI continues to function as the dominant conceptual and pedagogical reference for understanding, teaching, and securing network communication, while TCP/IP remains the model that actually describes the protocol suite running across the modern Internet. Understanding both — and the historical and technical reasons for their differences — remains essential for anyone studying or practising computer networking today [4][8][17].

References

[1] AWS, "What is the OSI Model? – 7 OSI Layers Explained," Amazon Web Services. Available: https://aws.amazon.com/what-is/osi-model/

[2] Webopedia, "What Are the 7 Layers of the OSI Model?" Available: https://www.webopedia.com/definitions/7-layers-of-osi-model/

[3] LIS Academy, "Exploring the ISO-OSI Reference Model: The 7-Layer Framework." Available: https://lis.academy/ict-fundamentals/iso-osi-reference-model-7-layer-framework/

[4] BMC Software, "What is the OSI Model? The 7 Layers Explained," BMC Blogs. Available: https://www.bmc.com/blogs/osi-model-7-layers/

[5] GeeksforGeeks, "OSI Model | 7 Layers Explained." Available: https://www.geeksforgeeks.org/computer-networks/open-systems-interconnection-model-osi/

[6] Corero Network Security, "What is the OSI Model? The 7 Layers Explained." Available: https://www.corero.com/what-is-the-osi-model/

[7] Imperva, "What is OSI Model | 7 Layers Explained." Available: https://www.imperva.com/learn/application-security/osi-model/

[8] Simple English Wikipedia, "TCP/IP model." Available: https://simple.wikipedia.org/wiki/TCP/IP_model

[9] Software AG Documentation, "History of TCP/IP." Available: https://documentation.softwareag.com/arf/4.1.1/en/webhelp/arf-webhelp/wca/admin/wtc_hist.htm

[10] "A Proposal for an Open Logistics Interconnection Reference Model for a Physical Internet," arXiv preprint, 2019. Available: https://arxiv.org/pdf/1904.05069

[11] BYJU'S, "TCP/IP Model." Available: https://byjus.com/govt-exams/tcp-ip-model/

[12] Computerworld, "Chapter 1: Network Overview." Available: https://www.computerworld.com/article/1551429/chapter-1-network-overview.html

[13] M. A. Padlipsky, "A Perspective on the ARPANET Reference Model," RFC 871, IETF, Sept. 1982. Available: https://tools.ietf.org/html/rfc871

[14] Tripwire, "An Overview of the OSI Model and Its Security Threats," State of Security. Available: https://www.tripwire.com/state-of-security/overview-osi-model-and-its-security-threats

[15] DestCert, "OSI Model Security Layers: Practical CISSP Study Guide for Modern Networks." Available: https://destcert.com/resources/network-security-osi-model/

[16] Abusix, "Network Security in the TCP/IP Model vs. OSI Model." Available: https://abusix.com/blog/network-security-in-the-tcp-ip-model-vs-osi-model/

[17] Exabeam, "OSI Layer Attacks: 38 Network Attacks by Layer and How to Mitigate." Available: https://www.exabeam.com/explainers/osi-layers/osi-layer-attacks-38-network-attacks-by-layer-and-how-to-mitigate/

[18] Codecademy, "Network Security Protocols: TLS, IPSec, and SSH Explained." Available: https://www.codecademy.com/article/network-security-protocols-tls-ipsec-ssh-explained

[19] I. Barona López et al., "Trends on Virtualisation with Software Defined Networking and Network Function Virtualisation," IET Networks, 2015. Available: https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-net.2014.0117

[20] B. R. Dawadi, D. B. Rawat, S. R. Joshi, and P. Manzoni, "Intelligent Approach to Network Device Migration Planning towards Software-Defined IPv6 Networks," Sensors, vol. 22, no. 1, 2022. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8747554/

[21] OnlineTutorialHub, "IPv6 Migration Explained: Transition From IPv4 To IPv6." Available: https://onlinetutorialhub.com/ccna/ipv6-migration-transition-ipv4-to-ipv6/

[22] B. R. Dawadi et al., "Software Defined IPv6 Network: A New Paradigm for Future Networking." Available: https://www.researchgate.net/publication/339354094_Software_Defined_IPv6_Network_A_New_Paradigm_for_Future_Networking

[23] Cloudswitch, "7 Layers Of OSI Model Encryption: From MACsec To HTTPS." Available: https://cloudswit.ch/blogs/osi-model-encryption-macsec-ipsec-tls-https/